

Week 3 Notes

STAT 251 SECTION 01

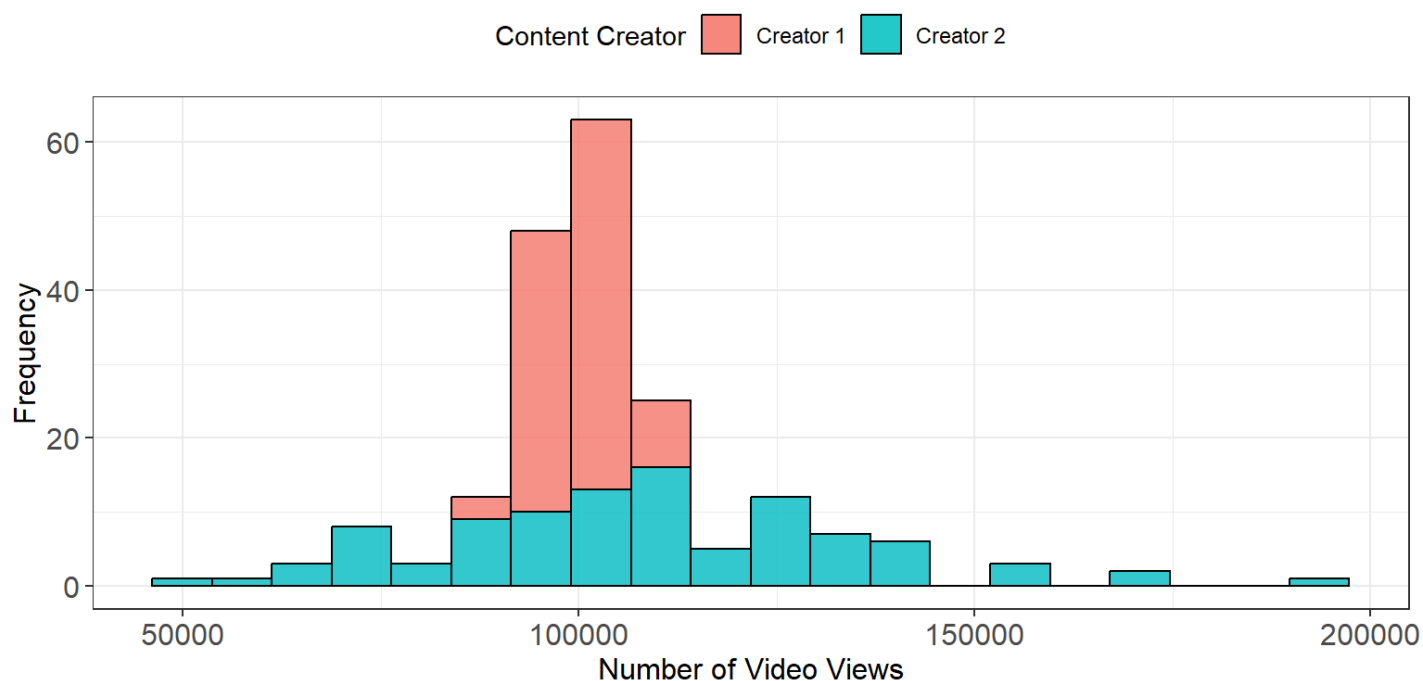
NO CLASS — Labor Day (Monday, September 7, 2026)

Describing spread

A description of a distribution is not complete without a measure of **spread** to sit alongside its center. Week 2 covered the measures of location and position — the mean, median and mode, and the percentiles and quartiles. This week asks a different question: how far do the observations fall from one another, and from the center? As with the measures of center, everything below applies to quantitative variables only.

Measures of spread

Even the simplest statistical description of a variable should include at least one measure of center and one measure of spread. The reason being that measures of center or location can be misleading if we do not consider how much the observations vary. Consider the following example which gives the distributions of the number of youtube video views of two content creators.



From this example we can see that both distributions have approximately the same mean. However, Content Creator 2 has considerably more variability in the number of views and, on a good day, can potentially get up to double his normal number of views.

In some sense giving several percentiles or the quartiles together tells us how different values are spread out across the distribution but there are other measures of spread that are preferred. One of the first and simplest measures we can use is the range. The **range** measures the distance between the largest and smallest observation and can be computed by subtracting the minimum value from the maximum value. Because the range uses the two most extreme points of the data it is extremely susceptible to outliers. For this reason, the range is not a measure of spread that gets used often.

A better measure of spread that can be computed from the first and third quartiles is the interquartile range or IQR. The **IQR** tells us how dispersed the inner 50% of the data.

- The IQR is defined as

$$IQR = Q3 - Q1$$

- It measures the distance between $Q1$ and $Q3$
- The IQR is very resistant to outliers since it ignores the extreme ends of a distribution

Take the 15 statistics exam scores from the Week 2 notes,

$$\{61, 61, 65, 65, 66, 68, 69, 73, 74, 75, 76, 78, 79, 90, 94\}$$

for which we found $Q1 = 65$, $Q2 = 73$ and $Q3 = 78$. Their interquartile range is

$$IQR = Q3 - Q1 = 78 - 65 = 13$$

so the middle half of the class scored within 13 points of one another. Compare that with the range, $94 - 61 = 33$: the range is more than twice as large, because it is measuring the distance out to the two most extreme students rather than the bulk of the class.

Deviations, variance and the standard deviation

The range and the IQR each describe spread using only two numbers out of the whole sample. A measure that uses **every** observation starts from the idea of a **deviation**: how far a single observation falls from the mean.

$$\text{deviation of } x_i = x_i - \bar{x}$$

An observation above the mean has a positive deviation, one below it has a negative deviation, and an observation sitting exactly at the mean has a deviation of zero.

- A natural first instinct is to average the deviations. That does not work, and it fails for a reason worth understanding: **the deviations always sum to zero**. The mean is the balancing point of the data, so the positive deviations cancel the negative ones exactly, for every dataset. Averaging them would report the same spread — zero — for data that is tightly packed and data that is wildly scattered.

Consider the ten die rolls $X = \{1, 2, 3, 3, 4, 4, 4, 5, 6, 6\}$, which have $\bar{x} = 3.8$:

x_i	$x_i - \bar{x}$	$(x_i - \bar{x})^2$
1	-2.8	7.84
2	-1.8	3.24
3	-0.8	0.64
3	-0.8	0.64
4	0.2	0.04
4	0.2	0.04
4	0.2	0.04
5	1.2	1.44
6	2.2	4.84
6	2.2	4.84

The deviation column sums to 0, as it must (R reports a value like 1.8e-15 rather than an exact 0 here, which is rounding error in the arithmetic, not a real difference). The fix is to remove the signs before averaging, and the standard choice is to **square** them. That gives the **variance**, written s^2 for a sample:

$$s^2 = \frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}$$

For the die rolls the squared deviations sum to 23.6, so $s^2 = 23.6/9 = 2.622$.

- Squaring has a side effect: the variance is in **squared units**. If the data are waiting times in minutes, the variance is in minutes squared, which is not a quantity anyone can picture. Taking

the square root undoes it and returns a number on the original scale. That is the **standard deviation**:

$$s = \sqrt{s^2} = \sqrt{\frac{\sum_{i=1}^n (x_i - \bar{x})^2}{n - 1}}$$

For the die rolls, $s = 1.619$ — back in “number of pips” rather than pips squared. Roughly speaking, s is the typical distance of an observation from the mean.

```
x <- c(1,2,3,3,4,4,4,5,6,6)
mean(x)
```

```
## [1] 3.8
```

```
var(x)    # s^2
```

```
## [1] 2.622222
```

```
sd(x)     # s
```

```
## [1] 1.619328
```

Why divide by $n - 1$ rather than n ?

Because the deviations are not n independent pieces of information. They are computed from $\bar{x}$, which was itself computed from the same data, and we already know they must sum to zero. Fix any $n - 1$ of the deviations and the last one is forced. So there are only $n - 1$ **degrees of freedom** among them, and dividing by $n - 1$ rather than n is what makes s^2 an unbiased estimate of the population variance σ^2 . Dividing by n would systematically underestimate it.

- Both s and s^2 are ≥ 0 , and both equal zero only when every observation is identical.
- Like the mean, both use every observation, so like the mean, **both are strongly influenced by outliers**. When a distribution is skewed or has outliers, the IQR is the more honest description of spread.

Which measures should you report?

A measure is **resistant** when a single extreme observation barely moves it. Whether a summary is resistant comes down to one thing: does it use every observation, or only positions in the ordered data?

Measure	Describes	Computed from	Resistant to outliers?
Mean $\bar{x}$	center	every observation	No
Median	center	the middle position	Yes
Mode	center	the most frequent value	Yes
Range	spread	the two most extreme values	No – the worst of all
IQR	spread	the two quartiles	Yes
Variance s^2	spread	every observation	No
Standard deviation s	spread	every observation	No

The practical consequence is that the measures come in **matched pairs**, and you should not mix them:

- If the distribution is roughly symmetric with no outliers, report the **mean and the standard deviation**.
- If it is skewed, or has outliers, report the **median and the IQR**.

A mean quoted next to an IQR answers two different questions and neither of them well.

A five number summary

Combining several numerical descriptors together is the best way to get a good feel for the characteristics of a distribution. A common method for summarizing a set of data is to use the **five number summary** which combines the minimum, maximum, and quartiles together.

- The five number summary: Minimum $Q1$ Median $Q3$ Maximum

Consider the five number summary of the statistics student exam scores:

The median tells us the center of the data, the quartiles show us the spread of the inner half of the data, the minimum and maximum tell us the total extent of spread in exam scores.

Often these five numbers are displayed graphically in a plot called a **box and whisker plot** (often called a boxplot). Together with the histogram, the boxplot is one of the most useful graphical summaries because it condenses a lot of information into an easy to create display.

- Steps for constructing a boxplot:
 - Mark the position of the first and third quartile and draw a box connecting them.
 - Draw a line inside of the box showing the position of the median
 - Work out how far $1.5 \times IQR$ reaches beyond each end of the box. These two positions, $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$, are the **fences**. They are not drawn.
 - The lower whisker is drawn out to the **smallest observation that is still at or above** the lower fence.
 - The upper whisker is drawn out to the **largest observation that is still at or below** the upper fence.
 - A whisker therefore ends on an actual data value, never on the fence itself. When no observation falls outside a fence, the whisker simply reaches the minimum or the maximum.
 - Any observation beyond a fence is marked as an outlier and drawn as its own point.

Identifying outliers: the $1.5 \times IQR$ rule

Up to now “outlier” has meant “a value that looks extreme”, which is a judgement call. The boxplot replaces it with an arithmetic rule. Define the two **fences**

$$\text{lower fence} = Q1 - 1.5 \times IQR \quad \text{upper fence} = Q3 + 1.5 \times IQR$$

and then:

- any observation **outside** a fence is flagged as an outlier and drawn as its own point;
- each whisker stops at the **most extreme observation still inside** the fences — which is why a whisker does not always reach the minimum or the maximum;
- the fences themselves are never drawn on the plot.

Being flagged does not make an observation wrong. It marks it as worth a second look — it may be a genuinely unusual case, or a recording error.

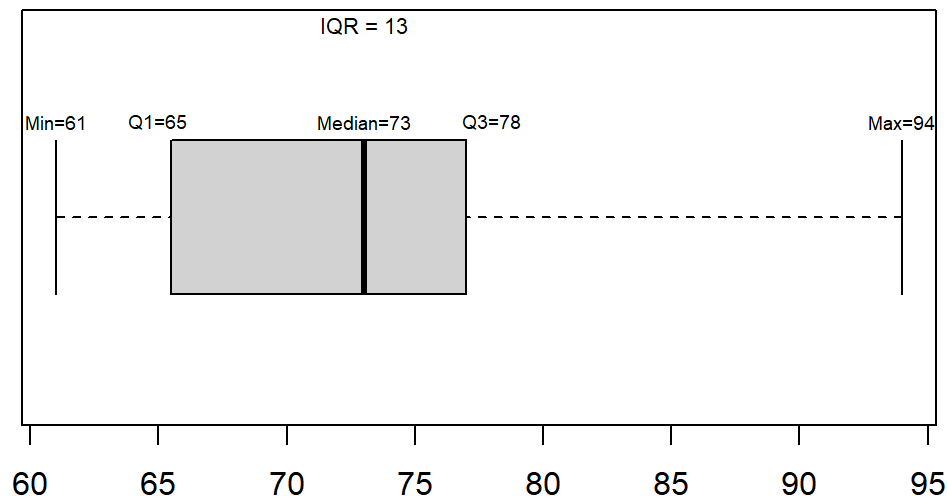
Applied to the 15 exam scores, where $Q1 = 65$, $Q3 = 78$ and $IQR = 13$:

$$65 - 1.5(13) = 45.5 \quad 78 + 1.5(13) = 97.5$$

Every score lies between 45.5 and 97.5, so nothing is flagged and the whiskers run all the way out to 61 and 94. The second example below is the interesting case, where they do not.

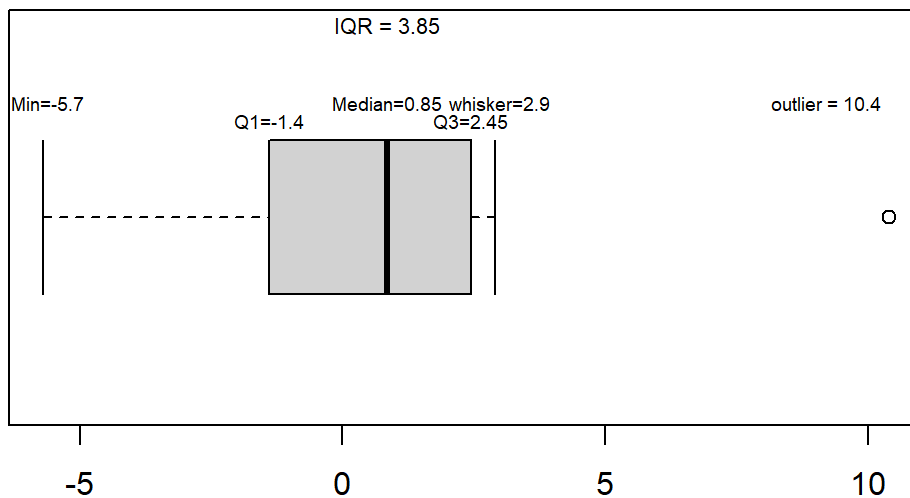
Example: consider the boxplot constructed for our example of student exam scores

$$\{61, 61, 65, 65, 66, 68, 69, 73, 74, 75, 76, 78, 79, 90, 94\}$$



Example2: consider the boxplot constructed for variable X

$$X = \{-5.7, -2.6, -1.5, -1.3, -0.4, 0.2, 1.5, 2.2, 2.3, 2.6, 2.9, 10.4\}$$

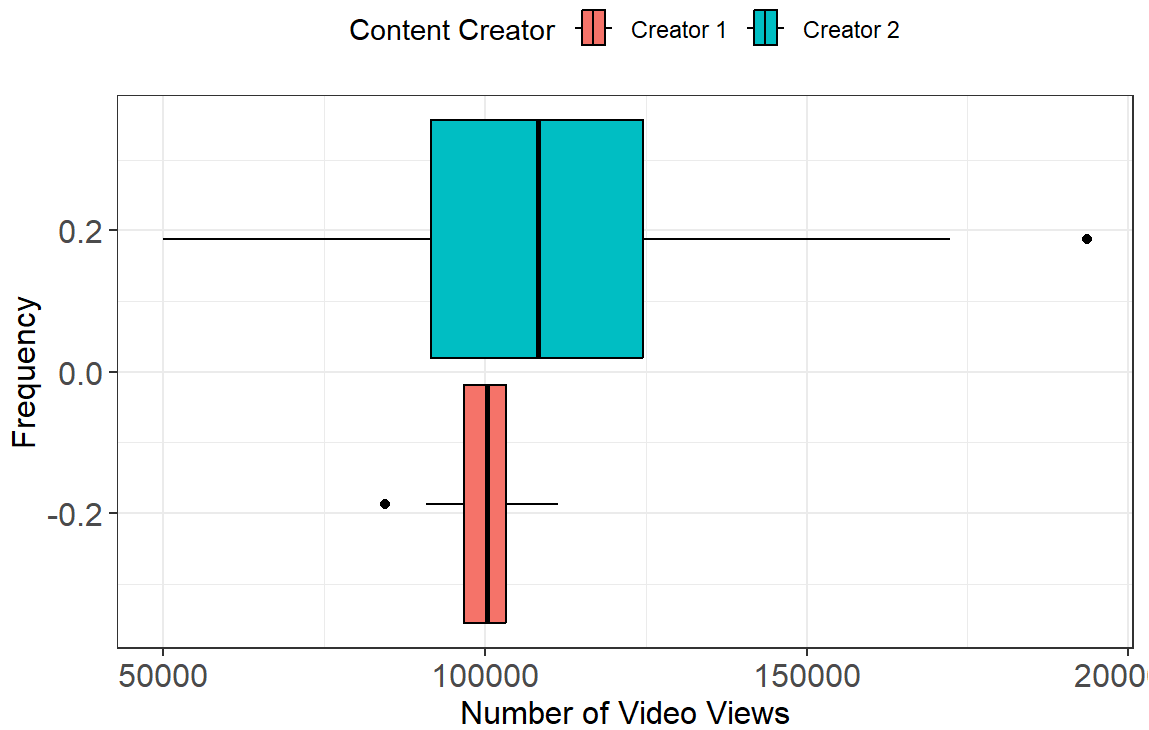


Here the rule bites. With $Q1 = -1.4$, $Q3 = 2.45$ and $IQR = 3.85$, the fences are

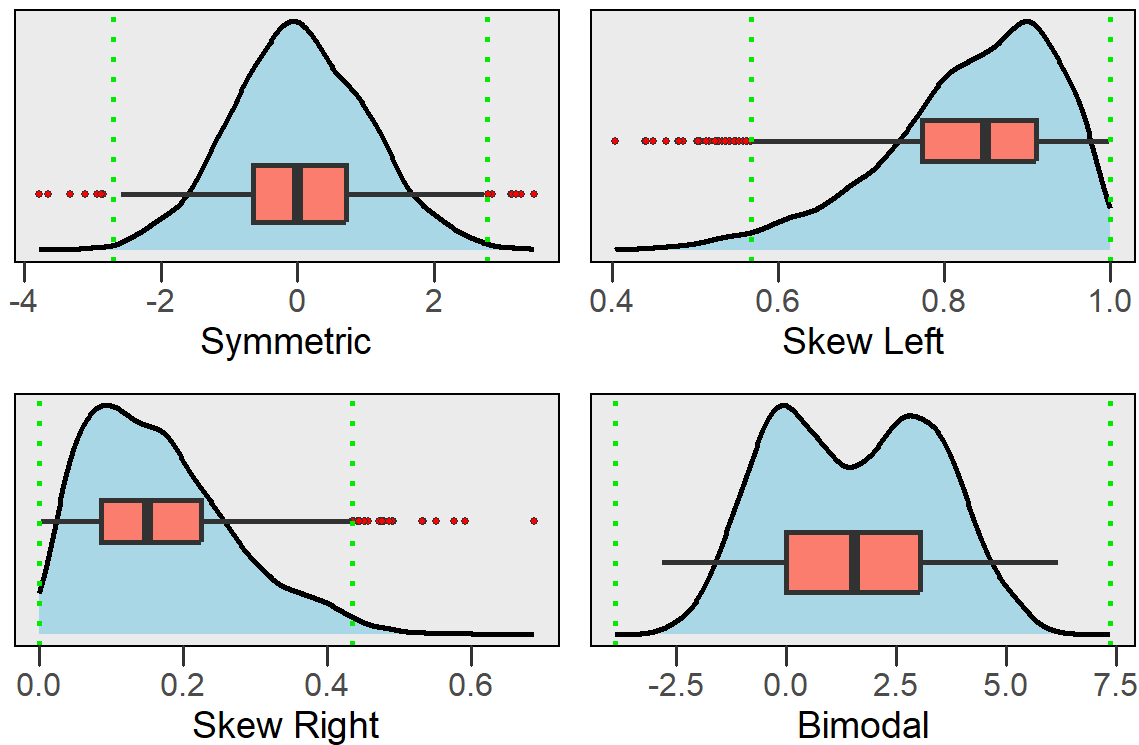
$$-1.4 - 1.5(3.85) = -7.175 \quad 2.45 + 1.5(3.85) = 8.225$$

The value 10.4 sits above the upper fence, so it is plotted as its own point and the upper whisker stops at 2.9, the largest observation that is not an outlier. Notice also how differently the two measures of spread describe this variable: the range is $10.4 - (-5.7) = 16.1$, inflated by the one extreme value, while the $IQR = 3.85$ is untouched by it.

- boxplots are great for comparing distributions and identifying skew. Consider our example for comparing the youtube views of two content creators.



- Using the boxplots makes the disparity in spread between the two Content Creators much more apparent. Boxplots are also great for identifying skew in a distribution. Consider the plot below which shows how you can interpret skew in relation to a boxplot.



- Notice from the plot above that the boxplot will not show if the distribution has multiple peaks or gaps. For this reason, a boxplot should always be plotted along with a histogram to ensure you capture all important features of a distribution.

Cumulative distributions

There is one more way to draw a distribution, and it is the one built for answering questions about **position**. The **cumulative distribution** shows the relationship between the value of a variable and the cumulative relative frequency. For discrete variables it is drawn as a **step function**: flat between observed values, jumping at each one by that value’s relative frequency. It only ever climbs, it starts at 0 and it finishes at 1.

Consider the sugar content, in grams per serving, of 20 US breakfast cereals.

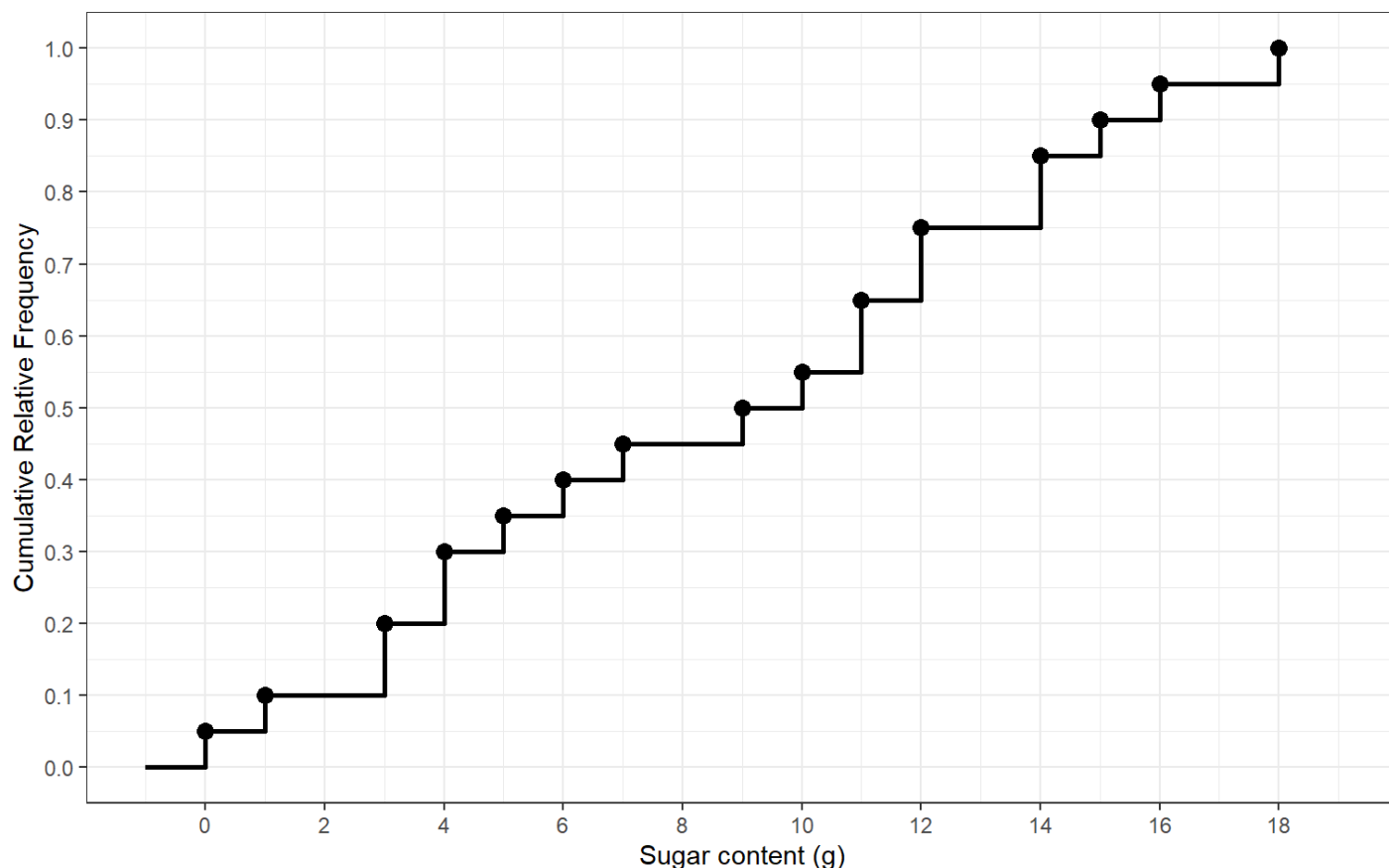
Cereal	Sugar (g)
Fiber One	0
Cheerios	1
Corn Flakes	3
Rice Krispies	3
Special K	4
Wheaties	4
All Bran	5
Life	6
Honey Bunches of Oats	7
Honey Nut Cheerios	9
Cinnamon Toast Crunch	10
Frosted Mini Wheats	11
Honeycomb	11
Captain Crunch	12
Frosted Flakes	12

Cereal	Sugar (g)
Apple Jacks	14
Froot Loops	14
Honey Smacks	15
Crackling Oat Bran	16
Raisin Bran	18

The frequency table carries a cumulative relative frequency column, which is all the cumulative distribution needs.

x	$F(x)$	$RF(x)$	$CRF(x)$
0	1	0.05	0.05
1	1	0.05	0.10
3	2	0.10	0.20
4	2	0.10	0.30
5	1	0.05	0.35
6	1	0.05	0.40
7	1	0.05	0.45
9	1	0.05	0.50
10	1	0.05	0.55
11	2	0.10	0.65
12	2	0.10	0.75
14	2	0.10	0.85
15	1	0.05	0.90
16	1	0.05	0.95
18	1	0.05	1.00

Plotting the cumulative relative frequency on the y -axis against the values of the variable on the x -axis gives the cumulative distribution.



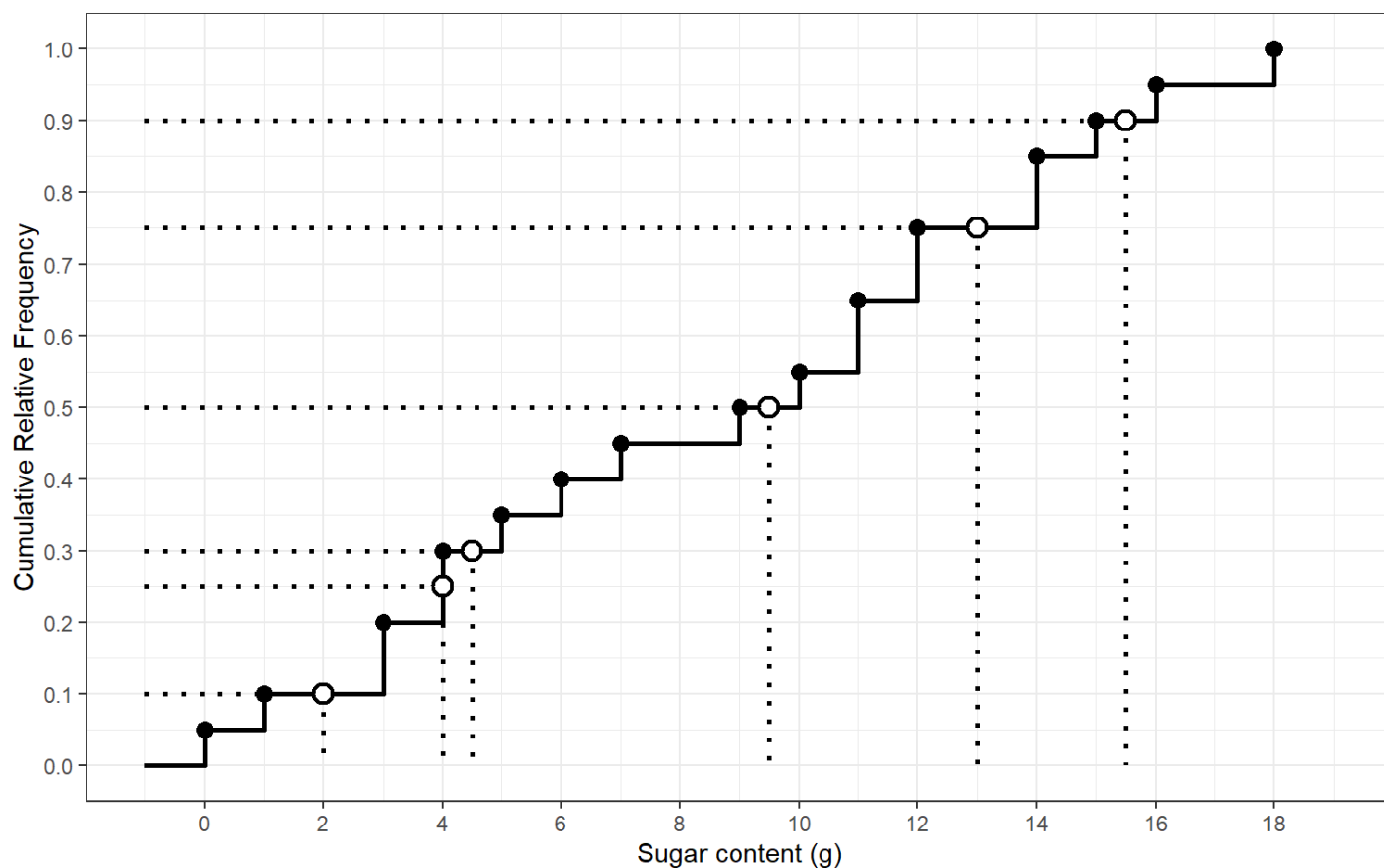
Reading percentiles off the cumulative distribution

This plot is a percentile machine. To find the p^{th} percentile: go up the y -axis to $p/100$, read across to the curve, and drop down to the x -axis. To go the other way — to ask what percentile a particular value sits at — start on the x -axis instead, go up to the curve, and read across to the y -axis.

One detail decides the answer whenever the reading lands on one of the flat runs:

- if reading across at $p/100$ **crosses a jump**, the p^{th} percentile is the value at that jump;
- if the horizontal bar sits **exactly** at height $p/100$, the p^{th} percentile is the **midpoint of that bar** — the average of the value it starts at and the next value up.

The second case only arises when n is even, which is why $n = 20$ is worth noticing here. Reading across at 0.90 lands on the bar that runs from 15 to 16, so the 90th percentile is $(15 + 16)/2 = 15.5$ and not 15.



Use the plot to confirm that the 10^{th} , 25^{th} , 30^{th} , 50^{th} , 75^{th} and 90^{th} percentiles are 2, 4, 4.5, 9.5, 13, 15.5 respectively.

The quartiles are just three of these percentiles, so the whole five number summary can be read off the same picture: the minimum is where the curve leaves 0 and the maximum is where it reaches 1.

$$Q1 = 4 \quad Q2 = 9.5 \quad Q3 = 13 \quad IQR = 13 - 4 = 9$$

These are the same three numbers you get by halving the ordered data, which is the point: the cumulative distribution is not a new definition of a quartile, only a faster way to read one off — and unlike the halving method it gives you *any* percentile, not just the quartiles.